Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

2.2 Let Titration Begin

2.2 Titration

Overview

2.2-1 Titration curve

2.2-2 Titration error

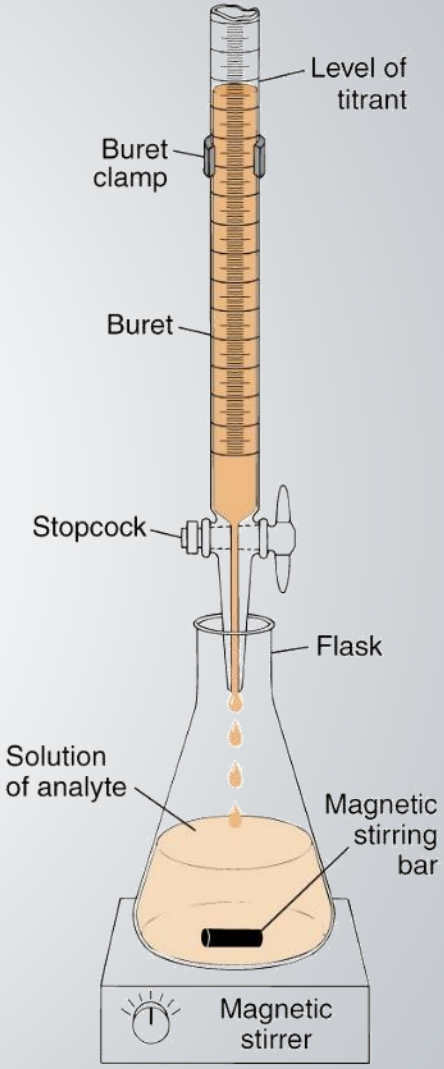
2.2-3 Find the end point

2.2-4 Titration types

Principles of titration

In a titration, increments of **titrant** are added to analyte until their reaction is complete. From the quantity of titrant required, the quantity of analyte can be calculated. (large *K* and proceed rapidly)

- Acid-base titration
- Precipitation titration
- Redox- titration



Typical setup for titration

Some concepts you need to know

Titration curve: a graph showing how the concentration of a reactant varies as titrant is added.

- ✓ To help to understand the chemistry that occurs during titrations
- ✓ To learn how experimental control can be exerted to influence the quality of an analytical titration

Equivalence point: point at which stoichiometric amounts of reactants (titrant and analyte) have been mixed.

End point: point near the equivalence point at which a sudden change in a physical property.

- Equivalence point is ideal (theoretical), what we actually measure is the end point

Some concepts you need to know

Titration error:

The difference between the end point and the equivalence point, it is inescapable.

- By choosing a physical property whose change is easily observed (such as the color change of an indicator)
- Blank titration, carry out the same procedure without analyte, which can decrease the titration error

Some concepts you need to know

V_e — volume of titrant at equivalence point

V_a — volume of titrant were added acid

V_b — volume of titrant were added base

2.2-1 Titration curve

2.2-1(A) Titration curve of <u>strong base</u> with strong acid

2.2-1(B) Titration curve of <u>weak acid</u> with strong base

2.2-1A Titration curve of strong base with strong acid

*Three regions for titration curve

$$OH^- + H^+ \overset{K_w}{\rightleftharpoons} H_2O$$

1. Before equivalence point, $0<V_a<V_e$

pH is determined by excess OH^-

$$[OH^-] = \frac{c(OH^-)\cdot V(OH^-) - c(H^+)\cdot V(H^+)}{V(H^+)+V(OH^-)}$$

2. At equivalence point, $V_a = V_e$ $Max.\frac{dpH}{dV_a}$

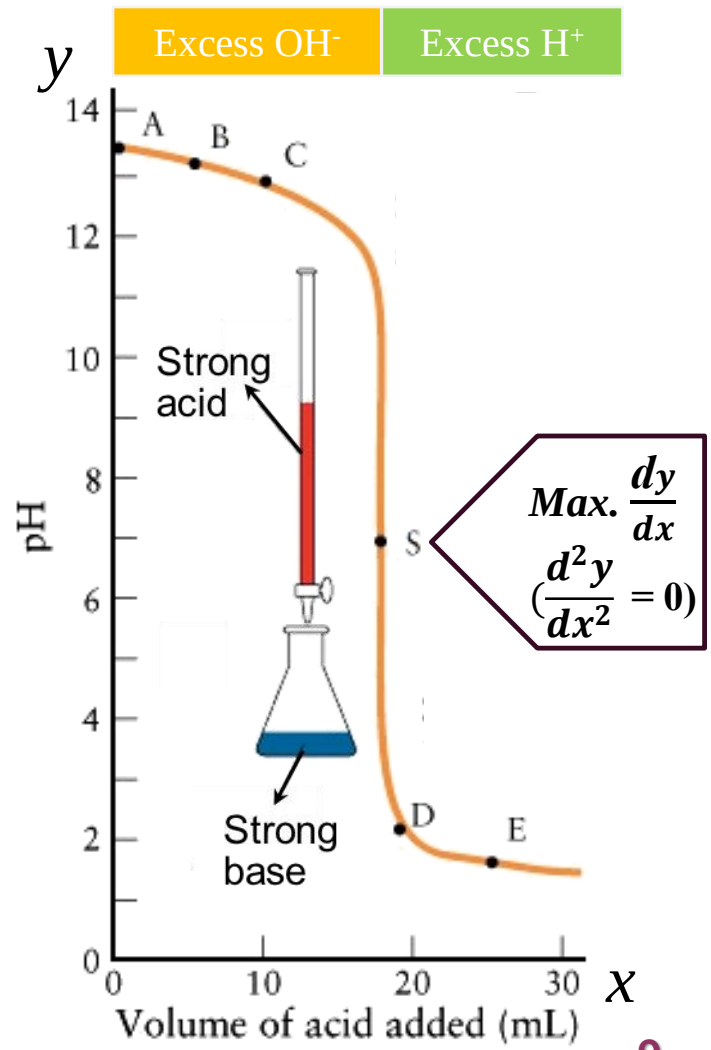
pH is determined by dissociation H_2O

$$[H^+]=[OH^-]=\sqrt{K_w}=10^{-7.00}$$

3. After equivalence point, $V_a > V_e$

pH is determined by excess H^+

$$[H^+] = \frac{c(H^+)\cdot V(H^+) - c(OH^-)\cdot V(OH^-)}{V(H^+)+V(OH^-)}$$



2.2-1A Titration of strong acid with strong base

What's the titration curve of strong acid with strong base?

Titration of 20.00mL 0.1000 mol·L^{-1} HCl with 0.1000 mol·L^{-1} NaOH

NaOH mL	*T*%	Excess HCl/mL	Excess NaOH/mL	pH	[H$^+$] calculation
0.00	0	20.00		1.00	Before titration[H$^+$] = *c*(HCl)
18.00	90.0	2.00		2.28	[H$^+$] = $\frac{c(H^+)\cdot V(H^+) - c(OH^-)\cdot V(OH^-)}{V(H^+)+V(OH^-)}$
19.80	99.0	0.20		3.30	
19.98	99.9	0.02		4.30	End point [H$^+$] = [OH$^-$] = 10$^{-7.00}$
20.00	100.0	0.00	0.00	7.00	
20.02	100.1		0.02	9.70	
20.20	101.0		0.20	10.70	[OH$^-$] = $\frac{c(OH^-)\cdot V(OH^-) - c(H^+)\cdot V(H^+)}{V(H^+)+V(OH^-)}$
22.00	110.0		2.00	11.68	
40.00	200.0		20.00	12.52	

2.2-1B Titration curve of weak acid with strong base

*Four regions for titration curve

1. Before any base added, $V_b = 0$

$$HA \overset{K_a}{\rightleftharpoons} H^+ + A^-$$

2. Before equivalence point, $0 < V_b < V_e$

$$pH = pKa + \log\frac{[A^-]}{[HA]}$$

$$pH = pKa, \text{ when } V_b = \frac{1}{2}V_e$$

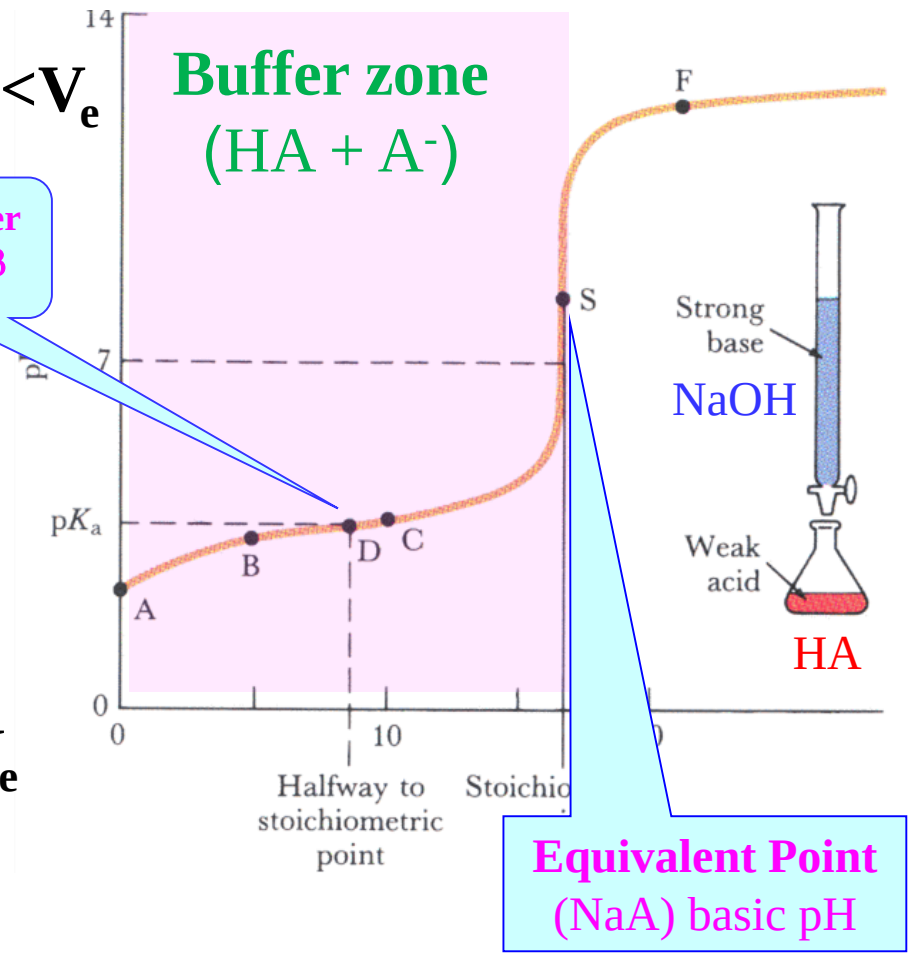
3. At equivalence point, $V_b = V_e$

$$A^- + H_2O \overset{K_b}{\rightleftharpoons} HA + OH^- \qquad K_b = \frac{K_w}{K_a}$$

4. After equivalence point, $V_b > V_e$

$$[OH^-] = c(NaOH) + [A^-]_{Hydrolysis}$$

$$HAc + NaOH \rightleftharpoons A^- + Na^+ + H_2O$$



Titration of 20.00mL 0.10 mol·L^{-1} HA/HAc with 0.10 mol·L^{-1} NaOH

K_a(HAc) = $10^{-4.76}$, K_a(HA) = $10^{-7.00}$)

NaOH mL	*T*%	compo-sition	pH (HAc)	pH (HA)	[H$^+$] calculation
0	0	HA	2.88	4.00	$[H^+] = \sqrt{K_a \cdot c_a}$
10.00	50.0	HA+A$^-$	4.76	7.00	when $f \geq 100\ K_a$, if not?
18.00	90.0	HA+A$^-$	5.71	7.95	$[H^+] = \frac{[HA]}{[A^-]} K_a$ (*buffer*)
19.80	99.0	HA+A$^-$	6.67	9.00	
19.96	99.8	HA+A$^-$	7.46	9.56	Henderson–Hasselbalch equ.
19.98	99.9	HA+A$^-$	7.76	9.70	
20.00	100.0	A$^-$	8.73	9.85	$[OH^-] = \sqrt{K_b \cdot c(A^-)}$
20.02	100.1	A$^-$+OH$^-$	9.70	10.00	when $c(A^-) \geq 100\ K_b$
20.04	100.2	A$^-$+OH$^-$	10.00	10.13	
20.20	101.0	A$^-$+OH$^-$	10.70	10.70	$[OH^-] = c(NaOH) + \cancel{[A^-]}$ Hydrolysis
22.00	110.0	A$^-$+OH$^-$	11.68	11.68	Same with titration of strong base

Titration of weak acid (HA) with strong base

At $V_b = V_e$

$\frac{dpH}{dV_b}$ is max., the curve is steepest

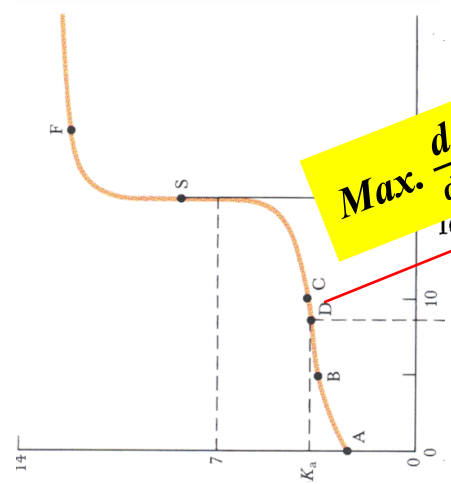
At $V_b = \frac{1}{2}V_e$, $pH = pKa$

The min. slope and max. β

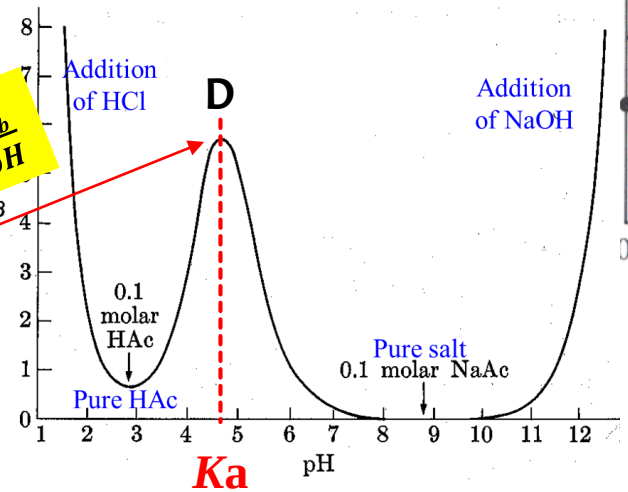
So, the buffer here is the most effective in resisting pH changes

$Ka(HAc) = 10^{-4.76}$

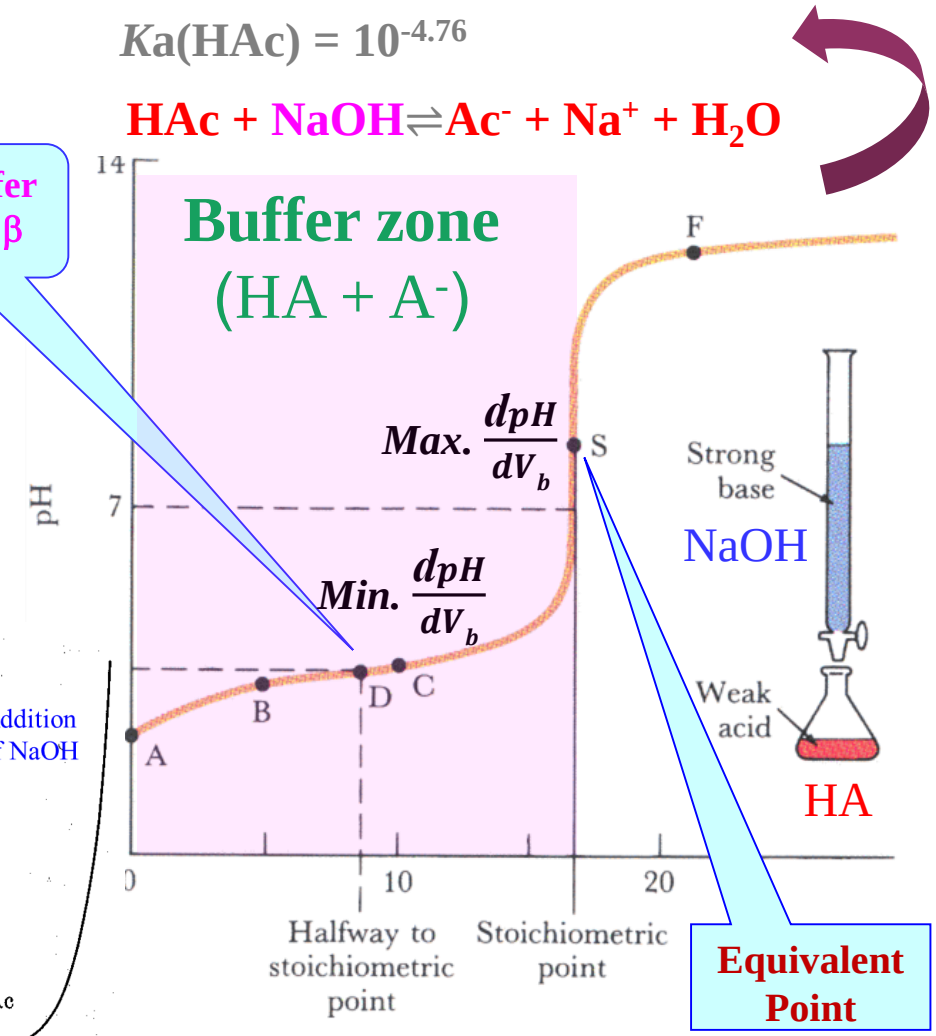
$HAc + NaOH \rightleftharpoons Ac^- + Na^+ + H_2O$



4.76

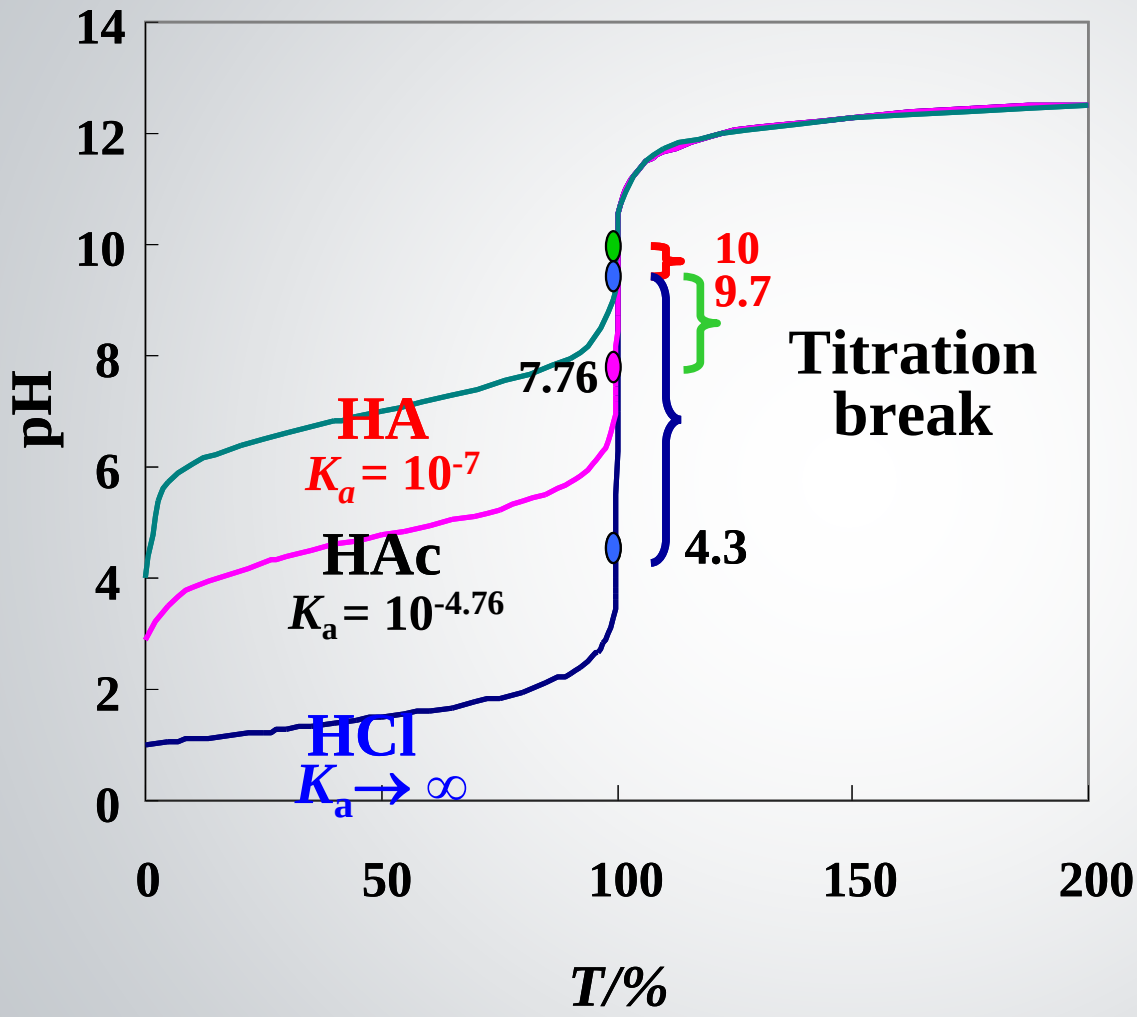


Ka
4.76



1. β increases with buffer [C]
2. $pH = pK_a$, $\beta_{max} = 0.575\ c$
3. Range: $pK_a - 1 < pH < pK_a + 1$

Titration of acid (different K_a) with strong base



0.10mol·L^{-1} NaOH

↓

0.10mol·L^{-1} Acid

Titration break

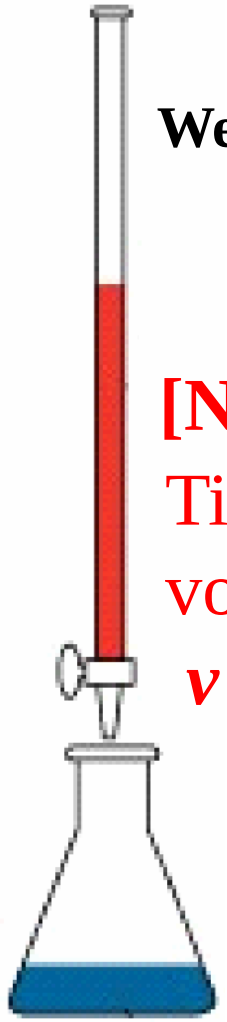
HCl: 4.3-9.7, pH=7

HAc: 7.76-9.7, pH=8.73

HA: 9.7-10.0, pH=9.85

Titration break is larger with increase K_a of acid.

2.2-2 Titration error



Titration function

Weak acid with strong base

[NaOH] = C

Titrated volume: ***v***

[HA] = C_0

Volume: $\mathbf{V_0}$

$$[HA]+[A^-]=\frac{C_0V_0}{v+V_0}$$ Mass balance

$$[HA]=\frac{C_0V_0}{v+V_0}-[A^-]$$

$$[Na^+]=\frac{Cv}{v+V_0}$$

$$[H^+][OH^-]=Kw$$ Equilibrium eq.

$$[H^+]+[Na^+]=[OH^-]+[A^-]$$ Charge balance

$$[H^+]+\frac{Cv}{v+V_0}=\frac{Kw}{[H^+]}+[A^-]$$

$$[A^-]=\frac{Cv}{v+V_0}+[H^+]-\frac{Kw}{[H^+]}$$

$$[HA] = \frac{C_0 V_0}{v + V_0} - [A^-]$$

$$[A^-] = \frac{Cv}{v + V_0} + [H^+] - \frac{Kw}{[H^+]}$$

$$\alpha_A \equiv \frac{[A^-]}{[HA] + [A^-]}$$

$$a_A = \frac{[A^-]}{\frac{C_0 V_0}{v + V_0} - [A^-] + [A^-]} = \frac{[A^-](v + V_0)}{C_0 V_0}$$

$$[A^-] = \alpha_A \frac{C_0 V_0}{v + V_0}$$

$$\alpha_A \frac{C_0 V_0}{v + V_0} = \frac{Cv}{v + V_0} \frac{C_0 V_0}{C_0 V_0} + [H^+] - \frac{Kw}{[H^+]} = \phi \frac{C_0 V_0}{v + V_0} + [H^+] - \frac{Kw}{[H^+]}$$

Titration function: the molar fraction titrated $\phi \equiv \frac{Cv}{C_0 V_0}$

At equivalent point $\phi = 1$

$$\phi = \{0 \cdots 1 \cdots\}$$

$$\alpha_A \frac{C_0V_0}{v+V_0} = \phi \frac{C_0V_0}{v+V_0} + [H^+] - \frac{Kw}{[H^+]}$$

$$\phi \frac{C_0V_0}{v+V_0} = \alpha_A \frac{C_0V_0}{v+V_0} - [H^+] + \frac{Kw}{[H^+]}$$

$$\phi = \alpha_A + \frac{v+V_0}{C_0V_0}\left[\frac{Kw}{[H^+]} - [H^+]\right]$$

$$\phi = 1 + \frac{v+V_0}{C_0V_0}\left[\frac{Kw}{[H^+]} - [H^+]\right]$$ Strong acid

$$V' = \frac{C_0 V_0}{C}$$ Volume titrated to the Equivalent-Point

$$V_{ep} = V_{end-point}$$ Volume titrated to the End-Point

Definition:

$$\text{Titration error} = \frac{V_{ep} - V'}{V'}$$

$$\text{Titration error} = \frac{V_{ep} - \frac{C_0 V_0}{C}}{\frac{C_0 V_0}{C}} = \frac{C V_{ep}}{C_0 V_0} - 1 = \phi_{ep} - 1$$

$$\phi = \frac{Cv}{C_0 V_0} \Rightarrow \phi_{ep} = \frac{C v_{ep}}{C_0 V_0}$$

Titration error = $\phi_{ep} - 1$

$$\phi = \alpha_A + \frac{v + V_0}{C_0 V_0}\left[\frac{Kw}{[H^+]} - [H^+]\right]$$

$$\phi_{ep} - 1 = (\alpha_A - 1) + \frac{v_{ep} + V_0}{C_0 V_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

Strong acid

$$\phi_{ep} - 1 = \frac{v_{ep} + V_0}{C_0 V_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

Approximation: The volume titrated to End-Point is close to the volume needed to Equivalent-Point.

$$v_{ep} \approx V'$$

$$\phi_{ep} - 1 \approx (\alpha_A - 1) + \frac{V' + V_0}{C_0 V_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

$$= (\alpha_A - 1) + \frac{\frac{C_0 V_0}{C} + V_0}{C_0 V_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

$$= (\alpha_A - 1) + \left[\frac{1}{C} + \frac{1}{C_0}\right]\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

$$= (\alpha_A - 1) + \frac{C + C_0}{C C_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

$$\text{Titration error} \approx (\alpha_A - 1) + \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

Strong acid $$\text{Titration error} \approx \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$

$$\alpha_A - 1 = \frac{[A^-]}{[HA]+[A^-]} - 1 = \frac{-[HA]}{[HA]+[A^-]} = -\frac{[H^+][A^-]}{Ka([HA]+[A^-])} \approx -\frac{[H^+]}{Ka}$$

Definition

Common denominator

At end-point [HA]<<[A-]

$$\boxed{\text{Titration error} = \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right] - \frac{[H^+]_{ep}}{Ka}}$$

$$\text{Titration error} = \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right] - \frac{[H^+]_{ep}}{Ka}$$

$$\text{Titration error} \approx \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right]$$ Strong acid

Conclusions:

(a) One can calculate the titration error from the *p*H at end-point (indicator)

In titration of strong acid with strong base, using indicator at *p*H=7, error = 0

(b) Titration error increases upon reagent dilution

Given C, C_0:

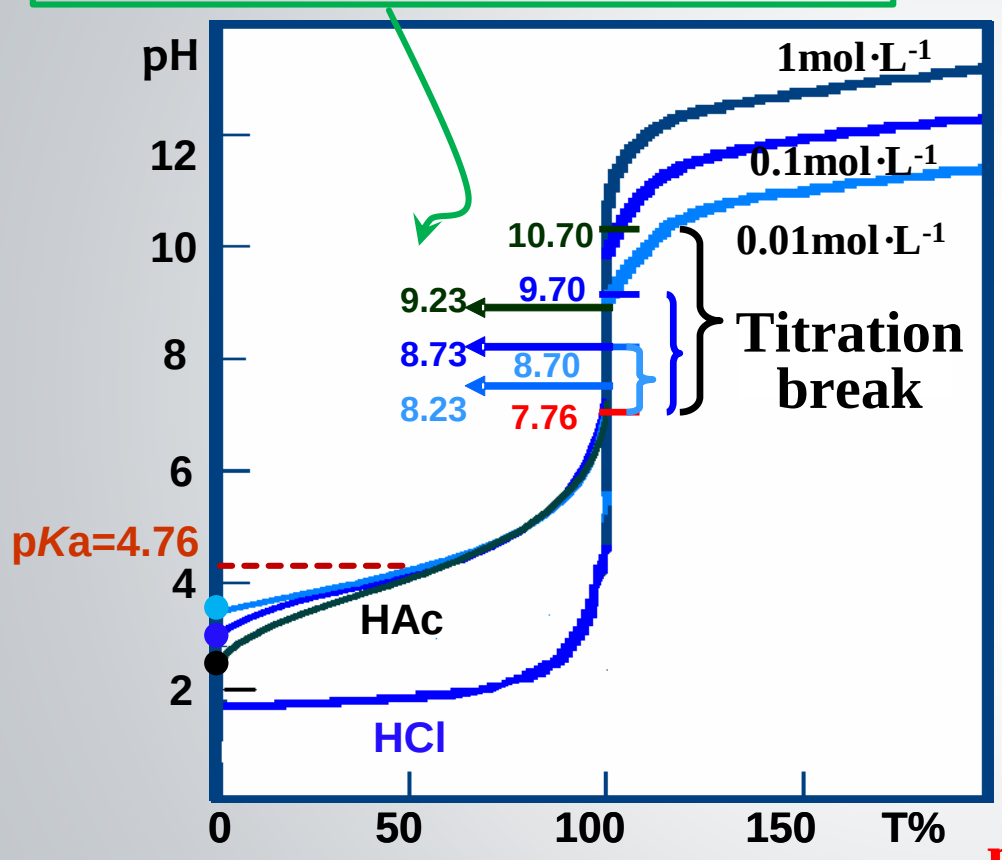
$$\frac{C + C_0}{CC_0} = x$$

Dilution by a factor of 2:

$$\frac{\frac{1}{2}C + \frac{1}{2}C_0}{\frac{1}{2}C\,\frac{1}{2}C_0} = 2x$$

Titration curve with diff. [HAc]

$$pH = 14 + \log\sqrt{k_b \cdot C(A^-)}$$



***K*a(HAc) = $10^{-4.76}$**

Titration break

0.01M: 7.76 – 8.70 by 0.01 M NaOH

0.10 M: 7.76 – 9.70 by 0.10 M NaOH

1.00 M: 7.76 – 10.70 by 1.00 M NaOH

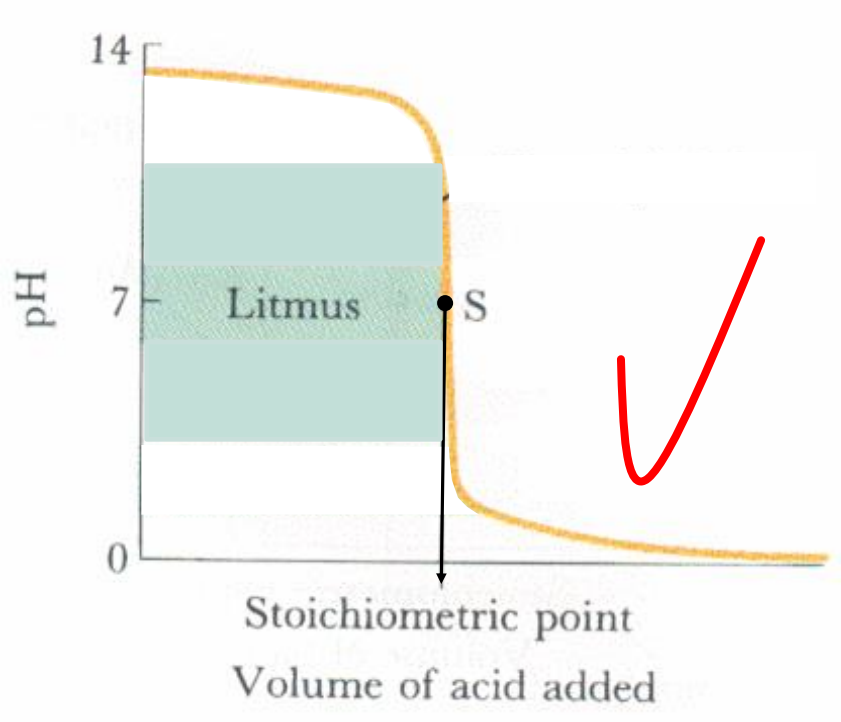
For [HAc] increase 10 times, the titration break becomes larger of one pH unit

pH = 14- pOH $[OH^-] = \sqrt{K_b \cdot c(A^-)}$

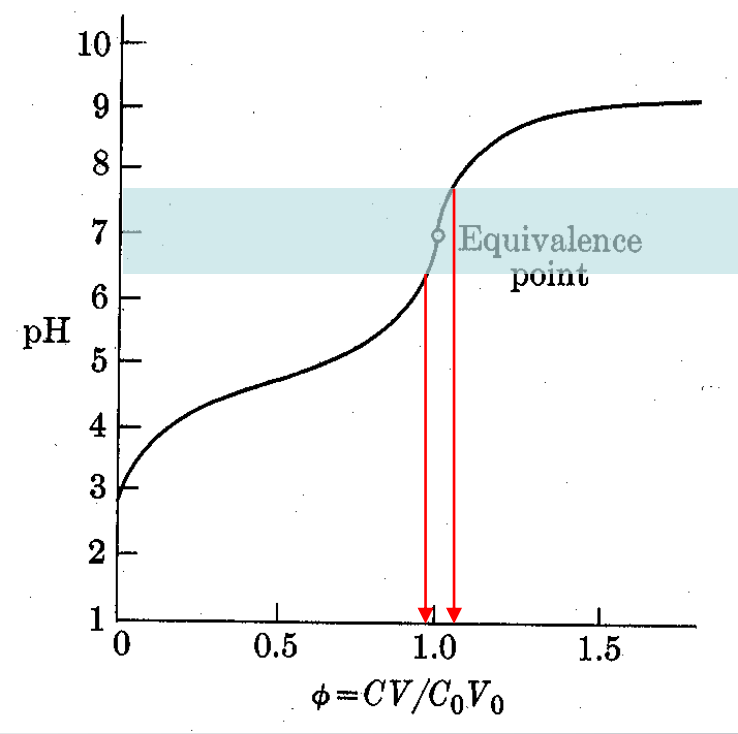
Titration break is larger at higher [C] of weak acid

Titration error resulting from the slope of the titration function at the equivalent-point

Titration of 25 mL0.1M NaOH with 0.1M HCl

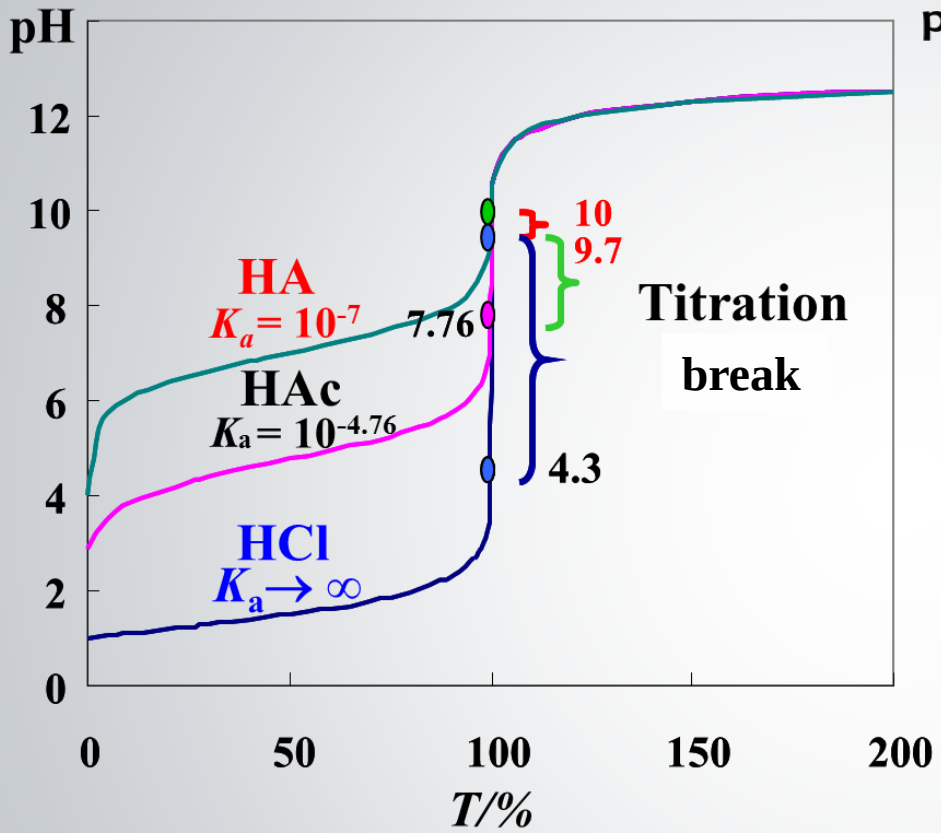


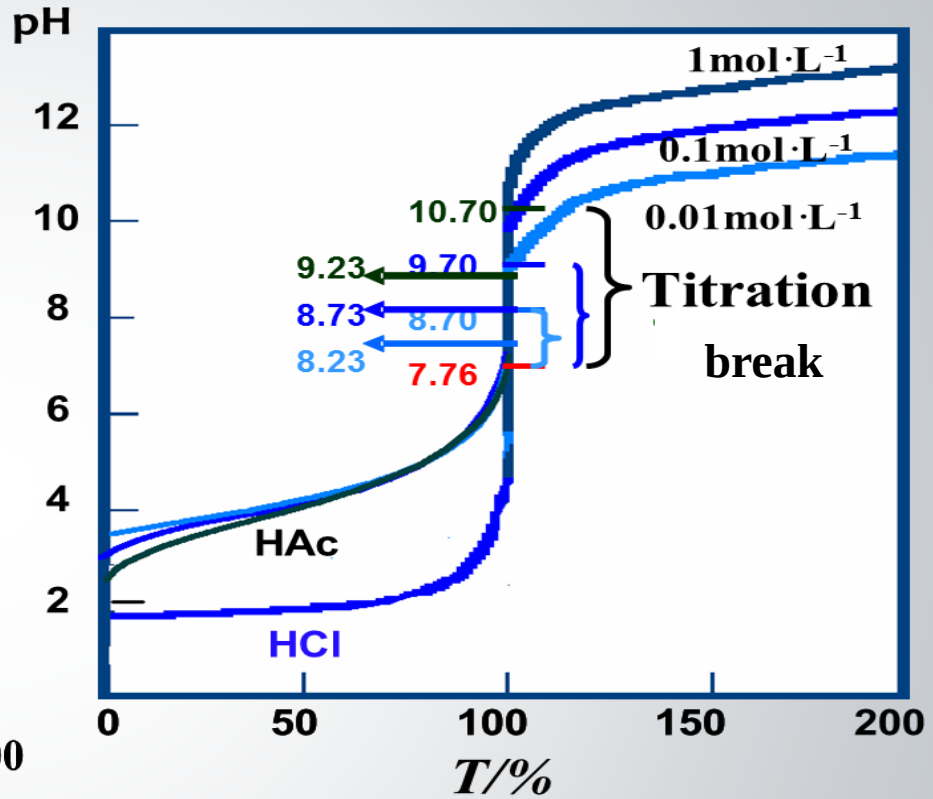
Titration of 25 mL 0.1M HAc with 0.1M NH_4OH



Which titration is more accurate?

Titration error resulting from *K*a and [C]





Titration break is larger, titration error would be less;

So, titration error is smaller under larger *K*a and higher [C].

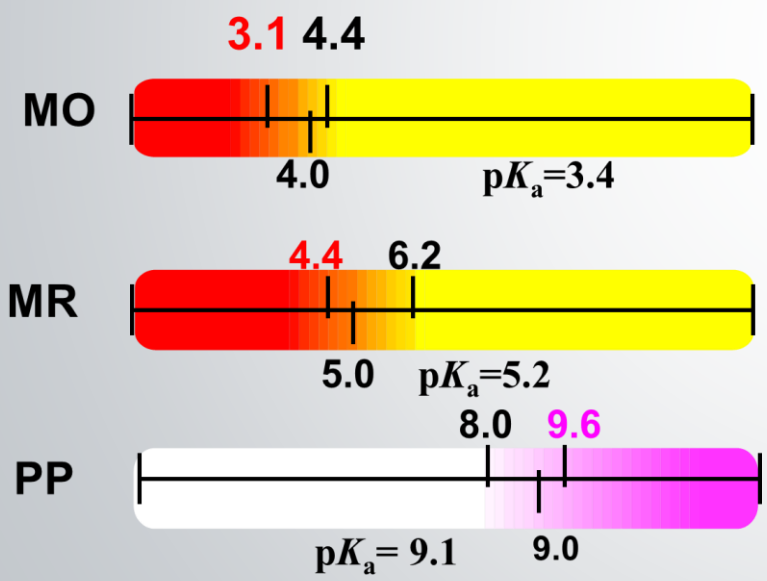
2.2-3 Find the end point

① By indicator

② By mathematics

Common Indicator for acid-base titration

Indicator	Color of acidic form	p*K*a	pH range of color change	Color of basic form
Methyl orange (MO)	red	3.4	3.2 to 4.4	yellow
Methyl red (MR)	red	5.0	4.8 to 6.0	yellow
Phenolphthalein (PP)	colorless	9.4	8.2 to 10.0	pink



$$pH = pKa + \log\frac{[A^-]}{[HA]}$$

How Indicator works?

$$HIn \rightleftharpoons H^+ + In^-$$

Color 1 Color 2

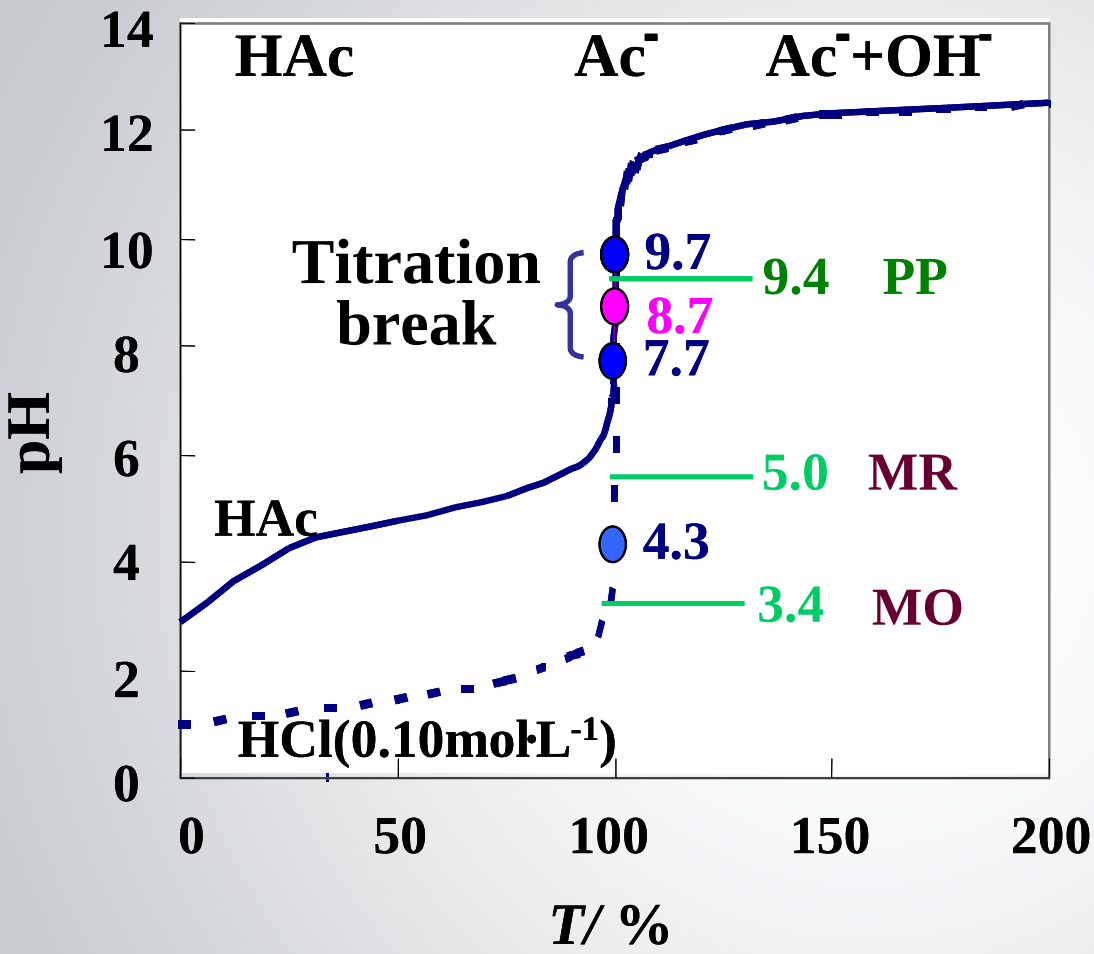
$$K_a = \frac{[H^+][In^-]}{[HIn]} \qquad \frac{[In^-]}{[HIn]} = \frac{K_a}{[H^+]}$$

[In-]/[HIn] > 10, pH > pK*a* + 1 → Basic Color

[In-]/[HIn] < 0.1, pH < pK*a* – 1→ Acidic color

Color change range: pH = p*Ka* ± 1

Titration of weak acid with strong base



T%---Titration function, molar fraction titrated

The indicator transition range should overlaps the steepest part of the titration curve

0.10mol·L^{-1}

NaOH

↓

0.10mol·L^{-1}

HAc $Ka(HAc) = 10^{-4.76}$

Choose indicator

MO (p*K*a=3.4)	3.2 to 4.4
MR (p*K*a=5.0)	4.8 to 6.0
PP (p*K*a=9.4)	8.2 to 10.0

Indicator for acid-base titration

Indicator	Color of acidic form	p*K*a	pH range of color change	Color of basic form
Thymol blue	red	1.7	1.2 to 2.8	yellow
Methyl orange (MO)	**red**	**3.4**	**3.2 to 4.4**	**yellow**
Bromophenol blue	yellow	3.9	3.0 to 4.6	blue
Bromophenol green	yellow	4.7	3.8 to 5.4	blue
Methyl red (MR)	**red**	**5.0**	**4.8 to 6.0**	**yellow**
Litmus	red	6.5	5.0 to 8.0	blue
Bromothymol blue	yellow	7.1	6.0 to 7.6	blue
Phenol red	yellow	7.9	6.6 to 8.0	red
Thymol blue	yellow	8.9	8.0 to 9.6	blue
Phenolphthalein (PP)	**colorless**	**9.4**	**8.2 to 10.0**	**pink**
Alizarin yellow R	yellow	11.2	10.1 to 12.0	red
Alizarin	red	11.7	11.0 to 12.4	purple

2.2-3 Find the end point

① By indicator

② By mathematics methods

a. First derivative (in Tutorial)

b. Gran plot (Leture 10)